

Module 12

Topics: Intro to graphs/charts/maps, Extra topic(s) TBD

Files to download: aidsvu_2016_newdx_county.sas7bdat, atlgrades.sas7bdat (from Module 5)
SAS Library files: SASHELP.CLASS, SASHELP.PRDSALE, MAPS.US, MAPS.COUNTIES

12.1 – Proc SGPLOT

PROC SGPLOT is a very useful procedure to generate graphs with SAS. Much like PROC REPORT, there are many books and articles on the topic and it takes time and practice to master all of the options. This section will provide you with an overview of the procedure to help you get started.

Let's do a few examples using the SASHELP dataset CLASS you are already familiar with from previous modules.

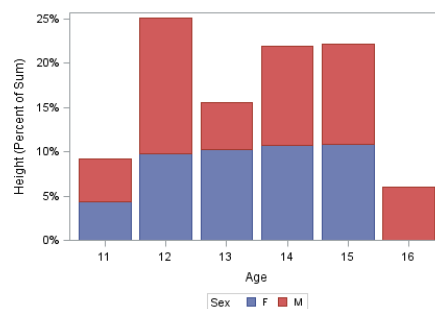
First, some general settings:

```
ods graphics on / reset=all;
ods graphics on / width=4.5in;
```

Now, for a graph.

```
proc sgplot data=sashelp.class pctllevel=graph;
  vbar age / response=height stat=percent
  group=sex;
run;
```

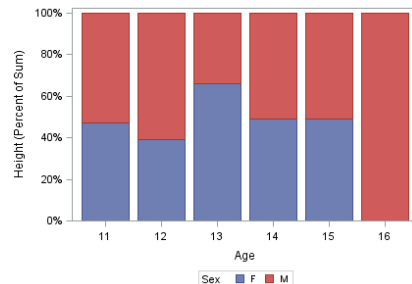
Notice that by specifying PCTLEVEL=GRAPH, the bars are all just portions of the total, and all together the bars add up to 100%.



If we change this to PCTLEVEL=GROUP, each bar will add up to 100%.

```
proc sgplot data=sashelp.class pctllevel=group;
  vbar age / response=height stat=percent
  group=sex;
run;
```

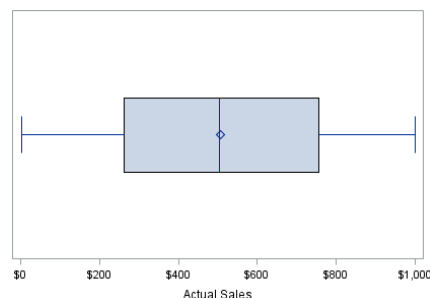
By changing to GROUP, the total is out of the category, which in this case is age. So we now have X% of those age 11 that are M and F, etc.



Another graph that is sometimes useful is the box plot. This plot shows the range of values as well as the median and outliers. We will use another SASHELP dataset, PRDSALE to demonstrate.

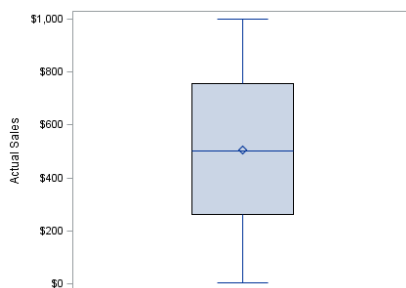
Box plots can be horizontal:

```
proc sgplot data=sashelp.prdsale;
  hbox actual;
run;
```



Or vertical:

```
proc sgplot data=sashelp.prdsale;
  vbox actual;
run;
```



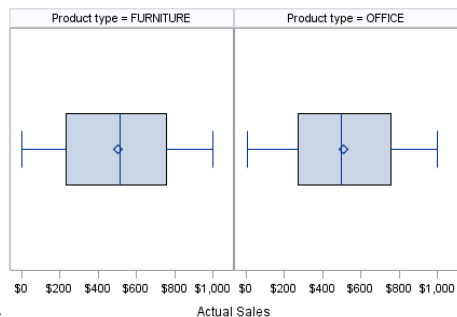
You can also use PROC SGPANEL to create these plots, both horizontal and vertical side by side. In this example we will have separate panels by product type (variable prodtype).

Horizontal:

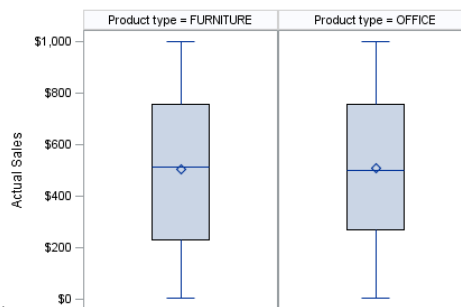
```
proc sgpanel data=sashelp.prdsale;
  panelby prodtype;
  hbox actual;
run;
```

Vertical:

```
proc sgpanel data=sashelp.prdsale;
  panelby prodtype;
  vbox actual;
run;
```



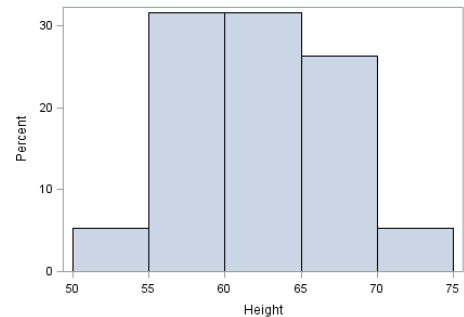
Horizontal:



Vertical:

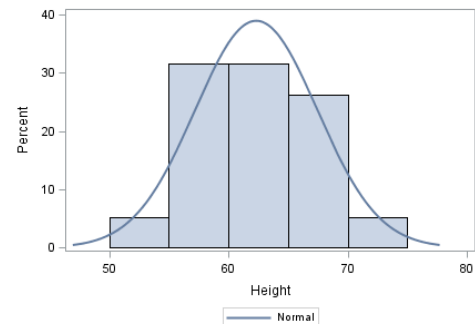
Histograms are also very useful.

```
proc sgplot data=sashelp.class;
  histogram height;
run;
```



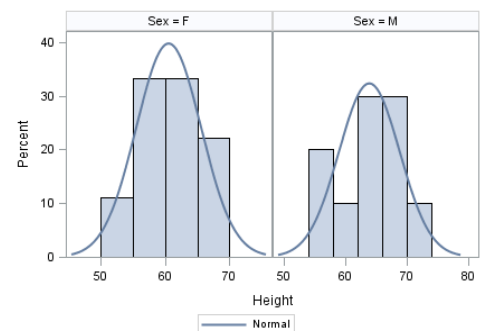
You can add a density plot to show the distribution.

```
proc sgplot data=sashelp.class;
  histogram height;
  density height;
run;
```



Both of these could also be run with SG PANEL to show graphs side by side.

```
proc sgpanel data=sashelp.class;
  panelby sex /
    uniscale=row;
  histogram height;
  density height;
run;
```



12.2 – Proc GMAP

PROC GMAP is also a very useful procedure. Again, there are many books and articles on the topic and it takes time and practice to master. This section will provide you with an overview of the procedure to help you get started.

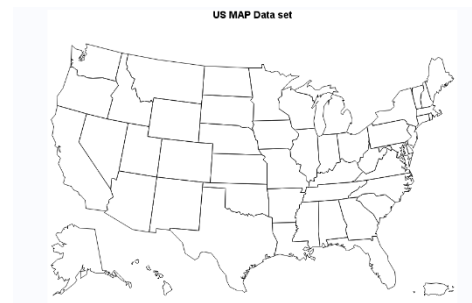
A choropleth map is produced by using a map data set (info about map boundaries) and a response data set (info to be displayed on the map such as data about the population). SAS has several map files available for use in the MAPS library.

Let's take a look at the US using a SAS supplied map file. To simply see an outline map, we can use the MAPS.US dataset as the map dataset and the response dataset. We will use v=e for the pattern because it is an empty pattern.

```

pattern v=e;
title 'US MAP Dataset';
proc gmap map=maps.us
    data=maps.us (obs=1) all;
    id state;
    choro state / nolegend;
run;
quit;

```



The FIPS code for Georgia is 13 – we can use this in a WHERE statement to limit the map to a specific area. First we have to limit the dataset to GA counties, then we can produce a map of the counties.

```

proc gproject data=maps.counties out=ga;
    where state = 13;
    id state;
run;

pattern v=e;
proc gmap data=ga (obs=1)
    map=ga all;
    id county;
    choro county / nolegend;
    note ' COUNTY MAP Dataset'
    j=l ' GA COUNTIES' ;
run;
quit;

```



Now let's try adding some real data to the mix. The dataset AIDS_VU_2016_NEWDX_COUNTY was pulled from www.aidsvu.org – it has the HIV new diagnoses rates and case counts for each county in GA. We will use this dataset as our response dataset and the GA dataset from the previous step as our map dataset. Note that this works because I have the county and state variables set up to match what is in the MAPS datasets.

First I am going to recreate the categories on aidsvu.org for the new diagnoses cases by creating a format.

```

proc format;
value newdx
    -1      = 'Supp'
    5       = '5'
    6       = '6'
    7-8     = '7 - 8'
    9-10    = '9 - 10'
    11-13   = '11 - 13'
    14-19   = '14 - 19'
    20-28   = '20 - 28'
    29-49   = '29 - 49'
    50-111  = '50 - 111'
    112-high = '112+'
;

```

```
run;
```

Next, I will use the following statement to reset SAS and make sure my colors will display properly.

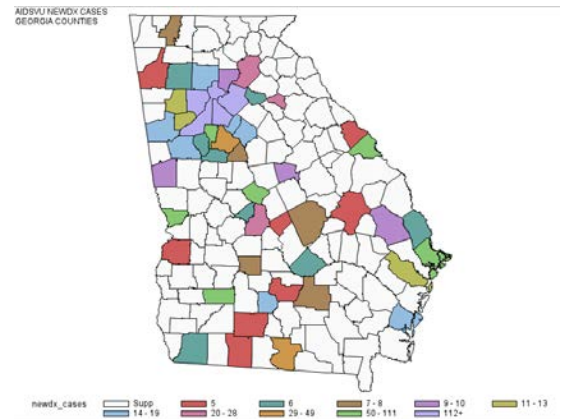
```
options reset=all border;
```

Now I am ready to run the GMAP procedure. Because the first category (newdx_cases = -1) means the value was suppressed, I will specify that the pattern for that is white. I will let SAS choose all the other colors, but I want to specify that the -1's are white since many of the counties are suppressed.

```
pattern1 v=ms c=grayfa;

proc gmap data=Aidsvu_2016_newdx_county
  map=ga;
  id county;
  choro newdx_cases / discrete;
  format newdx_cases newdx.;
  note ' AIDSVM NEWDX CASES'
  j=l ' GEORGIA COUNTIES' ;

run;
quit;
```



The following are a few great resources to get more familiar with the types of maps SAS can produce.

- This one goes through all the basics of PROC GMAP and gets into a lot of detail about annotating.
<http://www2.sas.com/proceedings/sugi29/251-29.pdf>
- This one has some neat examples making three-dimensional maps.
<http://www2.sas.com/proceedings/sugi27/p138-27.pdf>
- This one shows what you can do with PROC GCHART.
https://support.sas.com/sassamples/graphgallery/PROC_GCHART_Graph_Types_Charts_Bar.html
- This one covers all different types of maps/charts you didn't know SAS could make!
<http://robslink.com/SAS/democd1/aaaindex.htm>

MOD 12: USE SAS COMMENTS TO NUMBER YOUR WORK.

1. Using the ATLGRADES dataset from Module 5, create a graph of your choice to show the distribution of grades in the dataset.
2. Using the MAPS.US and MAPS.COUNTIES datasets as we did at the beginning of 12.2, try creating a county map outline for a state of your choice – it can be any state other than GA. Note that you will have to identify the FIPS code for your state so you can limit the counties in the map.